Exercise (4.1.1).

Proof. We are given a finite group G, a subgroup H, a representation (U, τ) of G, and a representation (V, π) of H. First, I claim that the map $\mathrm{Hom}_G(U, V^G) \to \mathrm{Hom}_H(U_H, V)$ given by:

$$\Phi \mapsto (u \mapsto \Phi(u)(e)$$

is a $\mathbb{C}$-linear transformation, where e denotes the identity of G. Linearity is clear, so we only need to show that the image of Φ is an H-equivariant map. For $h \in H$ and $u \in U$, we have:

$$\Phi(\tau(h)u)(e) = (\pi^G(h)\Phi(u))(e) = \Phi(u)(eh) = \Phi(u)(he) = \pi(h)\Phi(u)(e)$$

as desired.

Conversely, we claim that the map $\mathrm{Hom}_H(U_H, V) \to \mathrm{Hom}_G(U, V^G)$ given by:

$$\Psi \mapsto (u \mapsto (g \mapsto \Psi(\tau(g)u)))$$

is also $\mathbb{C}$-linear. Linearity is again clear, so we wish to show that the image of Ψ is G-equivariant and that this image evaluated at u is an element of V^G for each u. For both, it will be convenient for a fixed Ψ to define $f_u : G \to V$ by:

$$f_u(g) = \Psi(\tau(g)u)$$

for each $u \in U$. Then the map above can be defined by $\Psi \mapsto (u \mapsto f_u)$. To see that f_u is in V^G, note that for any $h \in H$ and $g \in G$,

$$f_u(hg) = \Psi(\tau(hg)u) = \Psi(\tau(h)(\tau(g)u)) = \pi(h)\Psi(\tau(g)u) = \pi(h)f_u(g)$$

as desired. Finally, we show that the overall map is G-equivariant; for $g, g' \in G$ and $u \in U$, we have:

$$f_{\tau(g')u}(g) = \Psi(\tau(g)(\tau(g')u)) = \Psi(\tau(gg')u) = f_u(gg') = (\pi^G(g')f_u)(g)$$

So that $f_{\tau(g')u} = \pi^G(g')f_u$ as desired.

Finally, to complete the proof, we show these two morphisms are inverses. For $\Phi \in \mathrm{Hom}_G(U, V^G)$, let Ψ be the image over the first morphism and Φ' be the image of Ψ across the second. Then, for $u \in U$ and $g \in G$,

$$\Phi'(u)(g) = \Psi(\tau(g)u) = \Phi(\tau(g)u)(e) = (\pi^G(g)\Phi(u))(e) = \Phi(u)(eg) = \Phi(u)(g)$$

so that $\Phi' = \Phi$. Conversely, for $\Psi \in \mathrm{Hom}_H(U_H, V)$, let Φ be the image over the second and Ψ' be the image of Φ over the first. Then, for $u \in U$,

$$\Psi'(u) = \Phi(u)(e) = \Psi(\tau(e)u) = \Psi(u)$$

so $\Psi' = \Psi$. Hence both compositions are the identity, and we have $\mathrm{Hom}_G(U, V^G) \cong \mathrm{Hom}_H(U_H, V)$ as $\mathbb{C}$-vector spaces.

Now let (π_1, V_1) and (π_2, V_2) be representations of the same finite group G with characters χ_1, χ_2. As per the hint, we know that both $\langle \chi_1, \chi_2 \rangle_G$ and $\dim \mathrm{Hom}_G(\pi_1, \pi_2)$ are linear in each coordinate. Namely, if π_1 is the direct sum of sub-representations, then χ_1 is the corresponding sum of characters, the inner product expands as a sum, the vector space of homs splits as a direct sum since each map can be defined on each summand, and the dimension adds. Similarly, using the projections, both terms are linear in π_2. So, showing these quantities are equal reduces to the case when π_1, π_2 are irreducible.

Let $S \in \mathrm{Hom}_G(\pi_1, \pi_2)$. Then note that $\ker S$ is a π_1-equivariant subspace of V_1, and $\mathrm{im}\, S$ is a π_2-equivariant subspace. Indeed, for $v \in \ker S$, $w = Sz \in \mathrm{im}\, S$, and $g \in G$ arbitrary:

$$S\pi_1(g)v = \pi_2(g)Sv = \pi_2(g)0 = 0$$
$$\pi_2(g)w = \pi_2(g)Sz = S\pi_1(g)z$$

so that $\pi_g(v) \in \ker S$ and $\pi_2(g)w \in \mathrm{im}\, S$. By irreducibility, this shows that the kernel and image are trivial. If either $\ker S = V_1$ or $\mathrm{im}\, S = 0$, then $S = 0$ is the zero map. Otherwise $\ker S = 0$ and $\mathrm{im}\, S = V_2$ and S is an isomorphism. So each nonzero element of $\mathrm{Hom}_G(\pi_1, \pi_2)$ is an isomorphism of representations.

We now have two cases. If $\pi_1 \ncong \pi_2$, then the above shows that $\mathrm{Hom}_G(\pi_1, \pi_2) = 0$, which has dimension zero. But in this case we also know $\langle \chi_1, \chi_2 \rangle = 0$, so the two quantities are equal as claimed. Otherwise $\pi_1 \cong \pi_2$, and we can pick an explicit isomorphim T. We know $\langle \chi_1, \chi_2 \rangle = 1$ in this case, so it suffices to show that $\mathrm{Hom}_G(\pi_1, \pi_2) = \mathbb{C}T$. To this end, let

$S \in \mathrm{Hom}_G(\pi_1, \pi_2)$. Either $S = 0 = 0T \in \mathbb{C}T$, or S is an isomorphism. Then $T^{-1}S$ is a π_1-equivariant automorphism of V_1. Let λ, v be an eigenvalue-eigenvector pair of $T^{-1}S$. Then

$$\mathrm{span}\{\pi_1(g)v \mid g \in G\}$$

is a nonzero π_1-invariant subspace of V_1 (since it contains v), so it must be all of V_1 by irreducibility. So, for $w \in V_1$, we can write:

$$w = \sum_i c_i \pi_1(g_i) v$$

for some finite collection of $c_i \in \mathbb{C}$ and $g_i \in G$. Then,

$$T^{-1}Sw = T^{-1}S\sum_i c_i\pi_1(g_i)v = \sum_i c_i(T^{-1}S\pi_1(g_i))v = \sum_i c_i(\pi_1(g)T^{-1}S)v = \sum_i c_i\pi_1(g_i)\lambda v = \lambda\sum_i c_i\pi_1(g_i)v = \lambda w$$

So that $T^{-1}S = \lambda I$, i.e. $S = \lambda T \in \mathbb{C}T$, completing the proof.

Finally, the last statement combines these two trivially:

$$\left\langle \sigma, \chi^G \right\rangle = \dim \mathrm{Hom}_G(\tau, \pi^G) = \dim \mathrm{Hom}_H(\tau_H, \pi) = \langle \sigma_H, \chi \rangle$$

as claimed. □

Exercise (4.1.2).

Proof. □

Exercise (4.1.3).

Proof. □

Exercise (4.1.4).

Proof. □

Exercise (4.1.5).

Proof. □

Exercise (4.1.6).

Proof. □

Exercise (4.1.7).

Proof. □

Exercise (4.1.8).

Proof. □

Exercise (4.1.9).

Proof. □

Exercise (4.1.10).

Proof. □

Exercise (4.1.11).

Proof. □

Exercise (4.1.12).

Proof. □

Exercise (4.1.13).

Proof. □

Exercise (4.1.14).

Proof. □

Exercise (4.1.15).

Proof. □

Exercise (4.1.16).

Proof. □

Exercise (4.1.17).

Proof. □

Exercise (4.1.18).

Proof. □

Exercise (4.1.19).

Proof. □

Exercise (4.1.20).

Proof. □